

Calc III Midterm 2 Review

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Contents

1	Definition Review	3
2	Theorem Review	7
3	Practice Problems	13
4	Answer Key	16

Chapter 1

Definition Review

Definition 1.1 (First order Taylor expansion). Let $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at $a \in U$, then

$$f(x) = f(a) + \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a)(x_i - a_i) + R_1(a, x)$$

where

$$\frac{R_1(a, x)}{\|x - a\|} \rightarrow 0 \text{ as } x \rightarrow a$$

Definition 1.2 (Second order Taylor expansion). Let $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable at $a \in U$, then

$$f(x) = f(a) + \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a)(x_i - a_i) + \frac{1}{2} \sum_{i,j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j}(a)(x_i - a_i)(x_j - a_j) + R_2(a, x)$$

where

$$\frac{R_2(a, x)}{\|x - a\|^2} \rightarrow 0 \text{ as } x \rightarrow a$$

Definition 1.3 (critical point). Let $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}$, a point $x_0 \in U$ is a **critical point** of f if either f is not differentiable at x_0 , or $Df(x_0) = 0$. A critical point that is not a local extremum is called a saddle point.

Definition 1.4 (quadratic function). A function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is called a **quadratic function** if it is given by

$$g(h_1, \dots, h_n) = \sum_{i,j=1}^n a_{ij} h_i h_j$$

where (a_{ij}) is an $n \times n$ matrix. We can also write g as follows:

$$g(h_1, \dots, h_n) = [h_1, \dots, h_n] \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix}$$

Definition 1.5 (Hessian matrix). Let $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}$, and suppose all the second-order partial derivatives $\frac{\partial^2 f}{\partial x_i \partial x_j}$ exist, then the Hessian matrix of f is the $n \times n$ matrix given by

$$Hf = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{bmatrix}$$

The Hessian as a quadratic function is defined by

$$Hf(x)(h) = \frac{1}{2} \begin{bmatrix} h_1 & \cdots & h_n \end{bmatrix} Hf(x) \begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix}$$

where $h = (h_1, \dots, h_n)$.

Definition 1.6 (degenerate/nondegenerate points). Let $f : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be of C^2 , let (x_0, y_0) be a critical point. We define the **discriminant**, $\mathcal{D}$, of the Hessian by

$$\mathcal{D} = \det(Hf) = \left(\frac{\partial^2 f}{\partial x^2} \right) \left(\frac{\partial^2 f}{\partial y^2} \right) - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2$$

If $\mathcal{D} \neq 0$, the critical point (x_0, y_0) is called **nondegenerate**; if $\mathcal{D} = 0$, the point (x_0, y_0) is called **degenerate**.

Definition 1.7 (positive, negative-definite). A quadratic function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is called **positive-definite** if $g(h) \geq 0$ for all $h \in \mathbb{R}^n$ and $g(h) = 0$ implies $h = 0$. Similarly, g is **negative-definite** if $g(h) \leq 0$ for all $h \in \mathbb{R}^n$ and $g(h) = 0$ implies $h = 0$.

Definition 1.8 (global extremum). Let $f : A \rightarrow \mathbb{R}$ be a function defined on $A \subset \mathbb{R}^2$ or $A \subset \mathbb{R}^3$. A point $x_0 \in A$ is said to be an **absolute maximum** if $f(x_0) \geq f(x)$ for all $x \in A$. Similarly, x_0 is an **absolute minimum** if $f(x_0) \leq f(x)$ for all $x \in A$.

Definition 1.9 (bounded set). A set $A \subset \mathbb{R}^n$ is said to be **bounded** if there is a number $M > 0$ such that $\|x\| \leq M$ for all $x \in A$.

Definition 1.10 (acceleration). Let $c(t)$ be a path, the **acceleration** $a(t)$ of $c(t)$ is

$$a(t) = c''(t)$$

Definition 1.11 (arc length). Let $c(t) = (x(t), y(t), z(t))$ be a path, then the length of the path in $\mathbb{R}^3$ from $t_0 \leq t \leq t_1$ is

$$\begin{aligned} L_{t_0 \rightarrow t_1}(c) &= \int_{t_0}^{t_1} (x'(t)^2 + y'(t)^2 + z'(t)^2)^{\frac{1}{2}} dt \\ &= \int_{t_0}^{t_1} \|c'(t)\| dt \end{aligned}$$

More generally, if $c(t) = (x_1(t), \dots, x_n(t))$ is a path in $\mathbb{R}^n$, then

$$L_{t_0 \rightarrow t_1}(c) = \int_{t_0}^{t_1} \left(\sum_{i=1}^n x_i'(t)^2 \right)^{\frac{1}{2}} dt$$

Definition 1.12 (vector field). A vector field is a function $F : A \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ that assigns $x \in \mathbb{R}^n$ to another vector $F(x) \in \mathbb{R}^n$.

Definition 1.13 (flow line). If F is a vector field, a **flow line** for F is a path $c(t)$ such that

$$c'(t) = F(c(t))$$

Intuitively speaking, flow lines are the “streamlines” threading through vector fields.

Definition 1.14 (divergence). Let F be a vector field in $\mathbb{R}^3$ $F = (F_1, F_2, F_3)$, the divergence of F is the **scalar field** (assigns one number to an given point (x, y, z)),

$$\operatorname{div} F := \nabla \cdot F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

More generally, if $F = (F_1, \dots, F_n)$ is a vector field on $\mathbb{R}^n$, its divergence is

$$\operatorname{div} F = \sum_{i=1}^n \frac{\partial F_i}{\partial x_i} = \frac{\partial F_1}{\partial x_1} + \dots + \frac{\partial F_n}{\partial x_n}$$

Definition 1.15 (curl). Let F be a vector field in $\mathbb{R}^3$, writing $F = (F_1, F_2, F_3)$, the **curl** of F is the vector field

$$\operatorname{curl} F := \nabla \times F = \det \begin{bmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{bmatrix}$$

If $\operatorname{curl} F = 0$, then we say the vector field is **irrotational**.

Definition 1.16. We say a region $D \subset \mathbb{R}^2$ is **y-simple** if there are continuous functions ϕ_1, ϕ_2 such that D is the set of points (x, y) satisfying

$$x \in [a, b], \quad \phi_1(x) \leq y \leq \phi_2(x)$$

Similarly, we define D to be **x-simple** if there are continuous ψ_1, ψ_2 such that D is the set of points (x, y) satisfying

$$y \in [c, d], \quad \psi_1(y) \leq x \leq \psi_2(y)$$

An **elementary** region is one that is x - or y -simple.

Definition 1.17 (injective, surjective). Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a map, we say T is **injective**, or **one-to-one**, on D^* , if for $x, y \in D^*$

$$Tx = Ty$$

implies

$$x = y.$$

We say T is **surjective**, or **onto** D , if for all $y \in D$, there exists x in the domain of T such that

$$Tx = y$$

If T is both injective and surjective, then we say T is **bijective**.

Definition 1.18 (Jacobian Determinant). Let $T : D^* \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be of C^1 defined by

$$T : \begin{pmatrix} u \\ v \end{pmatrix} \mapsto \begin{pmatrix} x(u, v) \\ y(u, v) \end{pmatrix}$$

The **Jacobian determinant** of T , denoted as $\frac{\partial(x, y)}{\partial(u, v)}$ is the determinant of the matrix $DT(u, v)$:

$$\frac{\partial(x, y)}{\partial(u, v)} = \det \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

Chapter 2

Theorem Review

Proposition 2.1 (extremums are critical points). Let $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable, where U is open. If x_0 is a local extremum, then $Df(x_0) = 0$.

Proposition 2.2 (extremum). Let $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be in C^3 , and x_0 is a critical point of f . If the Hessian $Hf(x_0)$ is positive-definite, then x_0 is a local minimum of f ; if $Hf(x_0)$ is negative-definite, then x_0 is a local maximum.

Proposition 2.3 (local minimum). Let $f(x, y)$ be of C^2 , and U is open in $\mathbb{R}^2$. A point (x_0, y_0) is a strict local **minimum** of f if the following conditions hold:

1.

$$\frac{\partial f}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = 0$$

2.

$$\mathcal{D}(x_0, y_0) > 0$$

where $\mathcal{D}$ is the **discriminant** of the Hessian, defined by

$$\mathcal{D} = \det(Hf) = \left(\frac{\partial^2 f}{\partial x^2} \right) \left(\frac{\partial^2 f}{\partial y^2} \right) - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2$$

where Hf is the 2×2 Hessian matrix.

3.

$$\frac{\partial^2 f}{\partial x^2}(x_0, y_0) > 0$$

Proposition 2.4 (local maximum). Let $f(x, y)$ be of C^2 , and U is open in $\mathbb{R}^2$. A point (x_0, y_0) is a strict local **maximum** of f if the following conditions hold:

1.

$$\frac{\partial f}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = 0$$

2.

$$\mathcal{D}(x_0, y_0) > 0$$

where $\mathcal{D}$ is the discriminant of the Hessian, defined above.

3.

$$\frac{\partial^2 f}{\partial x^2}(x_0, y_0) < 0$$

Proposition 2.5 (saddle points). Let $f(x, y) : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be of C^2 , if $\frac{\partial f}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = 0$, and $\mathcal{D}(x_0, y_0) < 0$, where $\mathcal{D}$ is the discriminant, then the critical point (x_0, y_0) is a saddle point, i.e., neither a maximum or a minimum.

Proposition 2.6 (continuous functions attain extremum on closed bounded sets). Let $f : D \rightarrow \mathbb{R}$ be continuous, where D is closed and bounded in $\mathbb{R}^n$. Then f assumes its absolute maximum and absolute minimum values at some point $x_0, x_1 \in D$.

Proposition 2.7. Let f be constrained to a surface S , if f has a max or a min at x_0 , then $\nabla f(x_0)$ is perpendicular to S at x_0 .

Proposition 2.8 (Lagrange). Suppose that $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : U \subset \mathbb{R}^n \rightarrow \mathbb{R}$ are C^1 functions. Let $x_0 \in U$ and $g(x_0) = c$, and let S be the level set for g at c , i.e., $S = \{x : g(x) = c\}$. Assume $\nabla g(x_0) \neq 0$, then if f has a local maximum or minimum on S at x_0 , then there exists some real number λ such that

$$\nabla f(x_0) = \lambda \nabla g(x_0)$$

Proposition 2.9 (Bordered Hessian). Let $f : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ and $g : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be smooth functions. Let $x_0 \in U$, $g(x_0) = c$, and let S be the level curve of g with value c . Assume that $\nabla g(x_0) \neq 0$ and that there exists a real number λ such that

$$\nabla f(x_0) = \lambda \nabla g(x_0)$$

Let $h = f - \lambda g$ and the bordered Hessian determinant is defined by

$$|\bar{H}| = \det \begin{vmatrix} 0 & -\frac{\partial g}{\partial x} & -\frac{\partial g}{\partial y} \\ -\frac{\partial g}{\partial x} & \frac{\partial^2 h}{\partial x^2} & \frac{\partial^2 h}{\partial x \partial y} \\ -\frac{\partial g}{\partial y} & \frac{\partial^2 h}{\partial x \partial y} & \frac{\partial^2 h}{\partial y^2} \end{vmatrix}$$

For f restricted to the curve S , at x_0 ,

1. If $|\bar{H}| > 0$, then x_0 is a local max.
2. If $|\bar{H}| < 0$, then x_0 is a local min.
3. If $|\bar{H}| = 0$, then it is inconclusive.

Proposition 2.10 (Newton's Second Law). Let F be the force acting on a particle of mass m , then

$$F = ma$$

where a is the acceleration.

Proposition 2.11 (gradient is irrotational). Let $f \in C^2$, viewing ∇f as a vector field, then

$$\nabla \times (\nabla f) = 0$$

Proposition 2.12 (divergence of a curl vanishes). For any C^2 vector field F ,

$$\nabla \cdot (\nabla \times F) = 0$$

Proposition 2.13 (Fubini's Theorem for rectangles). Let f be a continuous function on a rectangular domain $R = [a, b] \times [c, d]$, then

$$\int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

Proposition 2.14 (Fubini's Theorem for general regions). Suppose D is a set of points (x, y) such that $y \in [c, d]$ and $\psi_1(y) \leq x \leq \psi_2(y)$, and similarly for $x \in [a, b]$, $\varphi_1(x) \leq y \leq \varphi_2(x)$. If f is continuous on D , then

$$\iint_D f(x, y) dA = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy dx = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx dy$$

Proposition 2.15. We have the following identities regarding divergence and curl:

1. $\nabla(f + g) = \nabla f + \nabla g$.
2. $\nabla(cf) = c\nabla f$, for constant c .
3. $\nabla(fg) = f\nabla g + g\nabla f$.
4. $\nabla(f/g) = (g\nabla f - f\nabla g)/g^2$, at points x where $g(x) \neq 0$.
5. $\operatorname{div}(F + G) = \operatorname{div}F + \operatorname{div}G$.
6. $\operatorname{curl}(F + G) = \operatorname{curl}F + \operatorname{curl}G$.
7. $\operatorname{div}(fF) = f\operatorname{div}F + F \cdot \nabla f$.
8. $\operatorname{div}(F \times G) = G \cdot \operatorname{curl}F - F \cdot \operatorname{curl}G$.
9. $\operatorname{div}\operatorname{curl}F = 0$.
10. $\operatorname{curl}(fF) = f\operatorname{curl}F + \nabla f \times F$.
11. $\operatorname{curl}\nabla f = 0$.
12. $\nabla^2(fg) = f\nabla^2g + g\nabla^2f + 2(\nabla f \cdot \nabla g)$.
13. $\operatorname{div}(\nabla f \times \nabla g) = 0$.
14. $\operatorname{div}(f\nabla g - g\nabla f) = f\nabla^2g - g\nabla^2f$.

Proposition 2.16 (integrability). For different assumptions on f , we have the following integrability results:

1. Let f be continuous and defined on a closed rectangle R , then f is integrable over R .
2. Let $f : R \rightarrow \mathbb{R}$ be a bounded function on R and suppose the set of points where f is discontinuous lies on a finite union of graphs of continuous functions, then f is integrable over R .

Proposition 2.17 (Fubini's Theorem for rectangles). For different assumptions on f , we have the following Fubini's theorem results:

1. Let f be a continuous function on a rectangular domain $R = [a, b] \times [c, d]$, then

$$\int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy = \iint_R f(x, y) dA$$

2. Let f be bounded with domain $R = [a, b] \times [c, d]$ and the discontinuities of f lie on a finite union of graphs of continuous functions. If the integral $\int_a^b f dy$ exists for each $x \in [a, b]$, then

$$\int_a^b \left(\int_c^d f(x, y) dy \right) dx$$

exists and

$$\int_a^b \int_c^d f(x, y) dy dx = \iint_R f(x, y) dA$$

Similar results hold if $\int_a^b f dx$ exists for each $y \in [c, d]$. If both hold simultaneously, then

$$\int_a^b \left(\int_c^d f(x, y) dy \right) dx = \int_c^d \left(\int_a^b f(x, y) dx \right) dy = \iint_R f(x, y) dA$$

Proposition 2.18 (Fubini's Theorem for general regions). Suppose D is a set of points (x, y) such that $y \in [c, d]$ and $\psi_1(y) \leq x \leq \psi_2(y)$, and similarly for $x \in [a, b]$, $\varphi_1(x) \leq y \leq \varphi_2(x)$. If f is continuous on D , then

$$\iint_D f(x, y) dA = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy dx = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx dy$$

Proposition 2.19 (simple-regions). If D is a x -simple region with $y \in [c, d]$, $\psi_1(y) \leq x \leq \psi_2(y)$, and if f is continuous on D , then

$$\iint_D f(x, y) dA = \int_c^d \left(\int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx \right) dy$$

Similarly, if D is y -simple, then

$$\iint_D f(x, y) dA = \int_a^b \left(\int_{\phi_1(x)}^{\phi_2(x)} f(x, y) dy \right) dx$$

If D is simple, then the two expressions above are equal.

For example, the area of a x -simple region D can be computed as

$$\iint_D dA = \int_c^d \psi_2(y) - \psi_1(y) dy$$

Proposition 2.20. Let A be 2×2 matrix with $\det(A) \neq 0$ and let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear map $Tx = Ax$. Then T transforms parallelograms into parallelograms and vertices into vertices.

Proposition 2.21. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear map, i.e., there exists $n \times n$ matrix A such that $Tx = Ax$, then T is injective iff surjective iff $\det(A) \neq 0$.

Theorem 2.1 (change of variables formula). Let D, D^* be elementary regions in $\mathbb{R}^2$, suppose $T : D^* \rightarrow D$ is both one-to-one and onto. Then for any integral function $f : D \rightarrow \mathbb{R}$, the **change of variable formula** states

$$\iint_D f(x, y) dx dy = \iint_{D^*} f(x(u, v), y(u, v)) |\det(J)| du dv$$

where

$$\det(J) = \left| \frac{\partial(x, y)}{\partial(u, v)} \right|$$

is the Jacobian determinant.

Proposition 2.22 (change of variables-polar coordinates). As a corollary to the theorem above, we have the following change of variables formula for polar coordinates:

$$\iint_D f(x, y) dx dy = \iint_{D^*} f(r \cos \theta, r \sin \theta) r dr d\theta$$

Proposition 2.23 (change of variables-triple). Let W, W^* be elementary regions in $\mathbb{R}^3$, and suppose $T : W^* \rightarrow W$ is bijective. Then the change of variables formula for triple integrals states:

$$\iiint_W f(x, y, z) dx dy dz = \iiint_{W^*} f(x(u, v, w), y(u, v, w), z(u, v, w)) |\det(J)| du dv dw$$

where $\det(J)$ is the Jacobian determinant.

$$\det(J) = \det \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

Proposition 2.24 (change of variables-triple cylindrical). As a corollary to the above, we have the following change of variables formula for cylindrical coordinates:

$$\iiint_W f(x, y, z) dx dy dz = \iiint_{W^*} f(r \cos \theta, r \sin \theta, z) r dr d\theta dz$$

Recall cylinder coordinates is the following

$$x = r \cos \theta, y = r \sin \theta, z = z$$

Proposition 2.25 (change of variables-triple spherical). As a corollary to the above, we have the following change of variables formula for spherical coordinates:

$$\iiint_W f(x, y, z) dx dy dz = \iiint_{W^*} f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi d\rho d\theta d\phi$$

Recall the spherical coordinates is setting up the following

$$x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi$$

Chapter 3

Practice Problems

Problem 3.1. Compute the second-order Taylor polynomial of $f(x, y) = \cos(xy)$ centered at the point $(1, \pi/2)$.

- Problem 3.2.**
1. Write the number 120 as a sum of three numbers x, y, z so that the sum of the products $xy + yz + xz$ is maximal.
 2. Write the number 27 as a sum of three numbers x, y, z so that the product xyz is maximal.

Problem 3.3. Find the absolute maximum and minimum of the function $f(x, y) = \sin x + \cos y$ on the rectangle $0 \leq x \leq 2\pi, 0 \leq y \leq 2\pi$.

- Problem 3.4.**
1. Find the shortest distance between the point $(1, 0, -1)$ and the plane $2x + y + z = 0$.
 2. Find the shortest distance between the plane $x + y + z = 10$ and the unit sphere.
 3. Find the shortest distance between the curve $(\sin t, \cos t, t + 1)$ where $0 \leq t \leq 1$ and the origin.

Problem 3.5. Find the absolute maximum and minimum of $f(x, y) = x^2 + 2y^2$ on the unit disk $D = \{(x, y) : x^2 + y^2 \leq 1\}$.

Problem 3.6. Find the path c such that $c(0) = (0, 1, -1)$ and $c'(t) = (t, t^2, -e^t)$.

Problem 3.7. Find the length of the path

$$c(t) = (\sin t, \cos t, t)$$

where $0 \leq t \leq \pi$.

Problem 3.8. A path c is said to be parametrized by arc length $\|c'(s)\| = 1$.

1. Let c be a curve parametrized by arc length, show that the length of c for $a \leq s \leq b$ is $b - a$.
2. Parametrize the circle $\{(x, y) : x^2 + y^2 = 4\}$ by arc length.

Problem 3.9. Consider the vector field $F(x, y, z) = (x^2, z, xy^2)$. Can F be a gradient field?

Problem 3.10. Evaluate the following integral

$$\iint_D (xy + x + 1) dx dy$$

where D is the region enclosed by $y = -x^2 + 1$ and $y = x^2 - 7$.

Problem 3.11. Evaluate the following integrals

1.

$$\iint_D (xy + y + 1) dx dy$$

2.

$$\iint_D e^{\frac{x-y}{x+y}} dx dy$$

where D is the triangle with vertices $(0, 0)$, $(2, 0)$, $(1, 1)$.

Problem 3.12. Find the volume of the tetrahedron bounded by $x = 0$, $y = 0$, $z = 0$, $y + x + z = 1$.

Problem 3.13. Evaluate the following integral:

$$\int_0^1 \int_{\sqrt{y}}^1 e^{-x^3} dx dy$$

Problem 3.14. Find the volume of the truncated paraboloid bounded by $z = 1$, $z = 4$, $z = x^2 + y^2$.

Problem 3.15. Evaluate the following integral

$$\iiint_W x dx dy dz$$

where W is the truncated paraboloid $z = 1$, $z = 4$, $z = x^2 + y^2$.

Problem 3.16. Evaluate the following integral

$$\iiint_W z dx dy dz$$

where W is the pyramid with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(1, 1, 0)$, $(0, 0, 1)$.

Problem 3.17. Evaluate the following integral

$$\iiint_W (x^2 + y^2 + z) dx dy dz$$

where W is the cylinder $x^2 + y^2 \leq 2$, $1 \leq z \leq 2$.

Problem 3.18. Evaluate the following integral

$$\iint_D \frac{1}{\sqrt{x^2 + y^2}} dx dy$$

where D is the unit disk.

Problem 3.19. Evaluate the following integral

$$\iint_D (x - y)^2 e^{x+y} dx dy$$

where D is the region bounded by $x + y = 1$, $x + y = 4$, $x - y = -2$, $x - y = 1$.

Problem 3.20. Evaluate the following integral

$$\iint_D (x^2 + y^2) dx dy$$

where D is the region in the first quadrant bounded by $xy = 1$, $xy = 2$, $x^2 - y^2 = 1$, $x^2 - y^2 = 4$. Use the change of variables $u = x^2 - y^2$, $v = xy$.

Problem 3.21. Find the volume of the ellipsoid $E = \{(x, y, z) : x^2 + \frac{y^2}{4} + z^2 \leq 1\}$.

Problem 3.22. Evaluate the following integral

$$\iiint_W \frac{1}{(x^2 + y^2 + z^2)^{5/2}} dx dy dz$$

where W is the annulus bounded between $x^2 + y^2 + z^2 = 1$ and $x^2 + y^2 + z^2 = 4$.

Problem 3.23. Find the volume of the region bounded by $z \geq \sqrt{x^2 + y^2}$ and $z \leq 4 - \sqrt{x^2 + y^2}$.

Chapter 4

Answer Key

Problem 4.1. Compute the second-order Taylor polynomial of $f(x, y) = \cos(xy)$ centered at the point $(1, \pi/2)$.

Proof. The second-order Taylor polynomial is given by the formula

$$P(x, y) = f((1, \pi/2)) + f_x((1, \pi/2))(x - 1) + f_y((1, \pi/2))(y - \pi/2) + \frac{1}{2} (f_{xx}((1, \pi/2))(x - 1)^2 + 2f_{xy}(1, \pi/2)(x - 1)(y - \pi/2) + f_{yy}(1, \pi/2)(y - \pi/2)^2)$$

Thus we compute

$$f_x = -y \sin(xy), f_y = -x \sin(xy), f_{xx} = -y^2 \cos(xy), f_{xy} = -xy \cos(xy), f_{yy} = -x^2 \cos(xy)$$

Thus

$$P(x, y) = -\frac{\pi}{2}(x - 1) - (y - \pi/2) - (x - 1)(y - \pi/2)$$

□

- Problem 4.2.**
1. Write the number 120 as a sum of three numbers x, y, z so that the sum of the products $xy + yz + xz$ is maximal.
 2. Write the number 27 as a sum of three numbers x, y, z so that the product xyz is maximal.

Proof. 1. We want to maximize $f(x, y, z) = xy + yz + xz$ subject to the constraint $x + y + z = 120$. Using Lagrange, let $g(x, y, z) = x + y + z$, set

$$\nabla f = \lambda \nabla g \Rightarrow \begin{cases} y + z = \lambda \\ x + z = \lambda \\ x + y = \lambda \\ x + y + z = 120 \end{cases} \Rightarrow x = y = z = 40$$

2. We want to maximize $f(x, y, z) = xyz$ subject to the constraint $x + y + z = 27$. Similarly, we set $h(x, y, z) = x + y + z$. Then

$$\nabla f = \lambda \nabla h \Rightarrow \begin{cases} yz = \lambda \\ xz = \lambda \\ xy = \lambda \\ x + y + z = 27 \end{cases} \Rightarrow \begin{cases} (13.5, 13.5, 0) \\ (13.5, 0, 13.5) \\ (0, 13.5, 13.5) \\ (9, 9, 9) \end{cases}$$

It's clear that $(9, 9, 9)$ gives the maximal product.

□

Problem 4.3. Find the absolute maximum and minimum of the function $f(x, y) = \sin x + \cos y$ on the rectangle $0 \leq x \leq 2\pi, 0 \leq y \leq 2\pi$.

Proof. We first find the critical points in the interior of the rectangle

$$\nabla f = (\cos x, -\sin y) = (0, 0) \Rightarrow x_1 = (\pi/2, \pi), x_2 = (3\pi/2, \pi)$$

We see $f(x_1) = 0, f(x_2) = -2$. Now we check the boundaries:

1. $x = 0$: $f(x, y) = \cos y$ is maximized at $(0, 0), (0, 2\pi)$ with $f = 1$, minimized $(0, \pi)$ with $f = -1$.
2. $x = 2\pi$, similar to the above.
3. $y = 0$, $f(x, y) = \sin x$ is maximized at $(0, \pi/2)$ with $f = 2$, minimized at $(0, 3\pi/2)$ with $f = 0$.
4. $y = 2\pi$, similar to the above.

The absolute maximum is 2 and minimum is -2 .

□

Problem 4.4. 1. Find the shortest distance between the point $(1, 0, -1)$ and the plane $2x + y + z = 0$.
 2. Find the shortest distance between the plane $x + y + z = 10$ and the unit sphere.
 3. Find the shortest distance between the curve $(\sin t, \cos t, t + 1)$ where $0 \leq t \leq 1$ and the origin.

Proof. 1. We minimize the function $f(x, y, z) = (x-1)^2 + y^2 + (z+1)^2$, subject to the constraint $2x + y + z = 0$. Using Lagrange,

$$\nabla f = \lambda \nabla g \Rightarrow \begin{cases} 2x - 2 = 2\lambda \\ 2y = \lambda \\ 2z + 2 = \lambda \\ 2x + y + z = 0 \end{cases} \Rightarrow \left(\frac{2}{3}, -\frac{1}{6}, -\frac{7}{6} \right)$$

Thus the shortest distance is $\sqrt{1/9 + 1/36 + 1/36} = 1/\sqrt{6}$.

2. This is minimizing the distance between the plane and the center of the unit sphere, i.e., the origin. Thus we are minimizing $f(x, y, z) = x^2 + y^2 + z^2$ subject to the constraint $x + y + z = 10$. Using Lagrange,

$$\nabla f = \lambda \nabla g \Rightarrow \begin{cases} 2x = \lambda \\ 2y = \lambda \\ 2z = \lambda \\ x + y + z = 10 \end{cases} \Rightarrow (10/3, 10/3, 10/3)$$

Thus the distance between this point and the sphere (not the origin) is $\sqrt{100/3} - 1 = \frac{10-\sqrt{3}}{\sqrt{3}}$.

3. We minimize the function $f(t) = \sin^2 t + \cos^2 t + t^2 + 2t + 1 = (t+1)^2$. Thus it is minimized at $t = 0$, for $0 \leq t \leq 1$.

□

Problem 4.5. Find the absolute maximum and minimum of $f(x, y) = x^2 + 2y^2$ on the unit disk $D = \{(x, y) : x^2 + y^2 \leq 1\}$.

Proof. We first examine the critical points in the interior $\{(x, y) : x^2 + y^2 < 1\}$:

$$\nabla f = (2x, 4y) = (0, 0) \Rightarrow x_1 = (0, 0)$$

We see $f(x_1) = 0$. Now let's check the boundary. We will present two methods:

1. Parametrize the curve: let $x = \cos t, y = \sin t, 0 \leq t < 2\pi$. We have $f(t) = \cos^2 t + 2\sin^2 t$, Hence

$$f'(t) = -2\cos t \sin t + 4\sin t \cos t = 2\sin t \cos t = \sin(2t)$$

Setting $f'(t) = 0$ gives $t = 0, \pi/2, \pi, 3\pi/2$, hence the points $(\pm 1, 0), (0, \pm 1)$. We see maximum is achieved at $(0, \pm 1)$ with $f = 2$, and minimum is achieved at the critical point $x_1 = (0, 0)$ with $f = 0$.

2. Lagrange multipliers:

$$\nabla f = \lambda \nabla g \Rightarrow \begin{cases} 2x = \lambda 2x \\ 4y = \lambda 2y \\ x^2 + y^2 = 1 \end{cases} \Rightarrow (0, \pm 1), (\pm 1, 0)$$

Again we see maximum is achieved at $(0, \pm 1)$ with $f = 2$, and minimum is achieved at the critical point $x_1 = (0, 0)$ with $f = 0$.

□

Problem 4.6. Find the path c such that $c(0) = (0, 1, -1)$ and $c'(t) = (t, t^2, -e^t)$.

Proof. By taking the antiderivative and using $c(0) = (0, 1, -1)$, we find

$$c(t) = (t^2/2, t^3/3 + 1, -e^t)$$

□

Problem 4.7. Find the length of the path

$$c(t) = (\sin t, \cos t, t)$$

where $0 \leq t \leq \pi$.

Proof. Using the arc length formula, we have

$$\int_0^\pi \|c'(t)\| dt = \int_0^\pi 1 dt = \pi$$

□

Problem 4.8. A path c is said to be parametrized by arc length $\|c'(s)\| = 1$.

1. Let c be a curve parametrized by arc length, show that the length of c for $a \leq s \leq b$ is $b - a$.
2. Parametrize the circle $\{(x, y) : x^2 + y^2 = 4\}$ by arc length.

Proof. 1. Using the arc length formula, we see

$$\int_a^b \|c'(s)\| ds = \int_a^b 1 ds = b - a$$

2. We can parametrize as follows:

$$c(t) = \left(2 \cos \frac{t}{2}, 2 \sin \frac{t}{2} \right), 0 \leq t \leq 4\pi$$

□

Problem 4.9. Consider the vector field $F(x, y, z) = (x^2, z, xy^2)$. Can F be a gradient field?

Proof. We compute its curl:

$$\text{curl}(F) = \det \begin{bmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & z & xy^2 \end{bmatrix} = (2xy - 1, -y^2, 0)$$

which is not identically zero, hence F cannot be a gradient field.

□

Problem 4.10. Evaluate the following integral

$$\iint_D (xy + x + 1) dx dy$$

where D is the region enclosed by $y = -x^2 + 1$ and $y = x^2 - 7$.

Proof. This can be seen by drawing a graph: the region we are integration over is

$$-2 \leq x \leq 2, \quad x^2 - 7 \leq y \leq -x^2 + 1$$

Thus we integrate

$$\begin{aligned} \iint_D (xy + x + 1) dx dy &= \int_{-2}^2 \int_{x^2-7}^{-x^2+1} (xy + x + 1) dy dx \\ &= \int_{-2}^2 (4x^3 - 2x^2 - 16x + 8) dx \\ &= 64/3 \end{aligned}$$

□

Problem 4.11. Evaluate the following integrals

1.

$$\iint_D (xy + y + 1) dx dy$$

2.

$$\iint_D e^{\frac{x-y}{x+y}} dx dy$$

where D is the triangle with vertices $(0, 0)$, $(2, 0)$, $(1, 1)$.

Proof. 1. We see the region can be described by

$$0 \leq y \leq 1, \quad y \leq x \leq 2 - y$$

Hence

$$\iint_D (xy + y + 1) dx dy = \int_0^1 \int_y^{2-y} (xy + y + 1) dx dy = 5/3$$

2. Let $u = x + y, v = x - y$, using the change of variables formula,

$$\begin{aligned} \iint_D e^{v/u} |\det(J)| du dv &= \frac{1}{2} \int_0^2 \int_0^u e^{v/u} dv du \\ &= \frac{1}{2} \int_0^2 u(e - 1) du \\ &= e - 1 \end{aligned}$$

□

Problem 4.12. Find the volume of the tetrahedron bounded by $x = 0, y = 0, z = 0, y + x + z = 1$.

Proof. It's best if you draw a graph. We see the integral is given by

$$\int_0^1 \int_0^{1-x} (1 - x - y) dy dx = \int_0^1 \left(\frac{1}{2} - x + \frac{x^2}{2} \right) dx = 1/6$$

□

Problem 4.13. Evaluate the following integral:

$$\int_0^1 \int_{\sqrt{y}}^1 e^{-x^3} dx dy$$

Proof. We change the order of integration:

$$0 \leq x \leq 1, \quad 0 \leq y \leq x^2$$

Hence

$$\begin{aligned} \int_0^1 \int_{\sqrt{y}}^1 e^{-x^3} dx dy &= \int_0^1 \int_0^{x^2} e^{-x^3} dy dx \\ &= \int_0^1 x^2 e^{-x^3} dx \\ &= \frac{1}{3} \left(1 - \frac{1}{e} \right) \end{aligned}$$

□

Problem 4.14. Find the volume of the truncated paraboloid bounded by $z = 1$, $z = 4$, $z = x^2 + y^2$.

Proof. We use the cylindrical coordinates: $x = r \cos \theta$, $y = r \sin \theta$, $z = z$, then

$$\int_0^{2\pi} \int_1^4 \int_0^{\sqrt{z}} 1 \cdot r dr dz d\theta = \frac{15\pi}{2}$$

□

Problem 4.15. Evaluate the following integral

$$\iiint_W x dx dy dz$$

where W is the truncated paraboloid $z = 1$, $z = 4$, $z = x^2 + y^2$.

Proof. We can use the same bounds and cylindrical coordinates as above, hence we get

$$\iiint_W x dx dy dz = \int_0^{2\pi} \int_1^4 \int_0^{\sqrt{z}} r^2 \cos \theta dr dz d\theta = 0$$

□

Problem 4.16. Evaluate the following integral

$$\iiint_W z dx dy dz$$

where W is the pyramid with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(1, 1, 0)$, $(0, 0, 1)$.

Proof. We have

$$\int_0^1 \int_0^{1-z} \int_0^{1-z} z dx dy dz = \int_0^1 z(1-z)^2 dz = \frac{1}{12}$$

□

Problem 4.17. Evaluate the following integral

$$\iiint_W (x^2 + y^2 + z) dx dy dz$$

where W is the cylinder $x^2 + y^2 \leq 2$, $1 \leq z \leq 2$.

Proof. Using the cylindrical coordinates, we have

$$\begin{aligned} \int_1^2 \int_0^{\sqrt{2}} \int_0^{2\pi} (r^2 + z) r d\theta dr dz &= 2\pi \int_1^2 \int_0^{\sqrt{2}} (r^3 + rz) dr dz \\ &= 2\pi \int_1^2 (1 + z) dz \\ &= 5\pi \end{aligned}$$

□

Problem 4.18. Evaluate the following integral

$$\iint_D \frac{1}{\sqrt{x^2 + y^2}} dx dy$$

where D is the unit disk.

Proof. We use the polar coordinates:

$$\iint_D \frac{1}{\sqrt{x^2 + y^2}} dx dy = \int_0^1 \int_0^{2\pi} \frac{1}{r} \cdot r d\theta dr = 2\pi$$

□

Problem 4.19. Evaluate the following integral

$$\iint_D (x - y)^2 e^{x+y} dx dy$$

where D is the region bounded by $x + y = 1, x + y = 4, x - y = -2, x - y = 1$.

Proof. We do the change of variables $u = x + y, v = x - y$, then $|\det(J)| = 1/2$, thus

$$\iint_D (x - y)^2 e^{x+y} dx dy = \frac{1}{2} \int_1^4 \int_{-2}^1 v^2 e^u dv du = \frac{3}{2}(e^4 - e)$$

□

Problem 4.20. Evaluate the following integral

$$\iint_D (x^2 + y^2) dx dy$$

where D is the region in the first quadrant bounded by $xy = 1, xy = 2, x^2 - y^2 = 1, x^2 - y^2 = 4$. Use the change of variables $u = x^2 - y^2, v = xy$.

Proof. We first compute the Jacobian: it's easier to start with the inverse of the Jacobian, $\partial(u, v)/\partial(x, y)$,

$$\det \begin{bmatrix} 2x & -2y \\ y & x \end{bmatrix} = 2(x^2 + y^2)$$

Hence the Jacobian $\partial(x, y)/\partial(u, v)$ is

$$\frac{1}{2(x^2 + y^2)}$$

Thus

$$\iint_D (x^2 + y^2) dx dy = \int_1^2 \int_1^4 (x^2 + y^2) |\det(J)| du dv = \int_1^2 \int_1^4 \frac{1}{2} du dv = 3/2$$

□

Problem 4.21. Find the volume of the ellipsoid $E = \{(x, y, z) : x^2 + \frac{y^2}{4} + z^2 \leq 1\}$.

Proof. We make the following change of variables: $u = x, v = y/2, w = z$, then one can compute the Jacobian $\det(J) = 2$. Then the volume is given by

$$\iiint_B |\det(J)| du dv dw$$

where B is the unit ball. Using the fact that the unit ball has volume $4\pi/3$, we see the volume of the ellipsoid is $8\pi/3$. $\square$

Problem 4.22. Evaluate the following integral

$$\iiint_W \frac{1}{(x^2 + y^2 + z^2)^{5/2}} dx dy dz$$

where W is the annulus bounded between $x^2 + y^2 + z^2 = 1$ and $x^2 + y^2 + z^2 = 4$.

Proof. Using the spherical coordinates, we have

$$\iiint_W \frac{1}{(x^2 + y^2 + z^2)^{5/2}} dx dy dz = \int_0^{2\pi} \int_0^\pi \int_1^2 \frac{1}{\rho^5} \rho^2 \sin \phi d\rho d\phi d\theta = 3\pi/2$$

$\square$

Problem 4.23. Find the volume of the region bounded by $z \geq \sqrt{x^2 + y^2}$ and $z \leq 4 - \sqrt{x^2 + y^2}$.

Proof. We use cylindrical coordinates

$$\int_0^{2\pi} \int_0^2 \int_r^{4-r} r dz dr d\theta = 16\pi/3$$

$\square$